

Area of Technical Uncertainty (ATU) in antimicrobial susceptibility testing (8 February 2024)

The Area of Technical Uncertainty (ATU) is a term coined by EUCAST to warn laboratories of uncertain interpretation of antimicrobial susceptibility test (AST) results. These may be by MIC determination or disk diffusion. The need for an ATU with MIC breakpoints is mostly related to the breakpoint dividing prominent MIC distributions – this could be significant overlap between susceptible and resistant parts of the distribution, or because of the appearance of an important resistant population which was not initially identified. The need for ATUs in disk diffusion is for the same reasons but sometimes because of a less than perfect correlation between MIC and inhibition zone diameters.

The ATU is normally defined by one specific MIC-value or by one or a range of inhibition zone diameters. The ATU is not a susceptibility test category like S, I and R and it does not interfere with the interpretation of results. It should not be reported as part of the clinical susceptibility report – however, on occasion when an uncertainty could not be resolved it is appropriate to inform the clinical colleague of the problem and to explain why a susceptibility report of S, I or R is not possible.

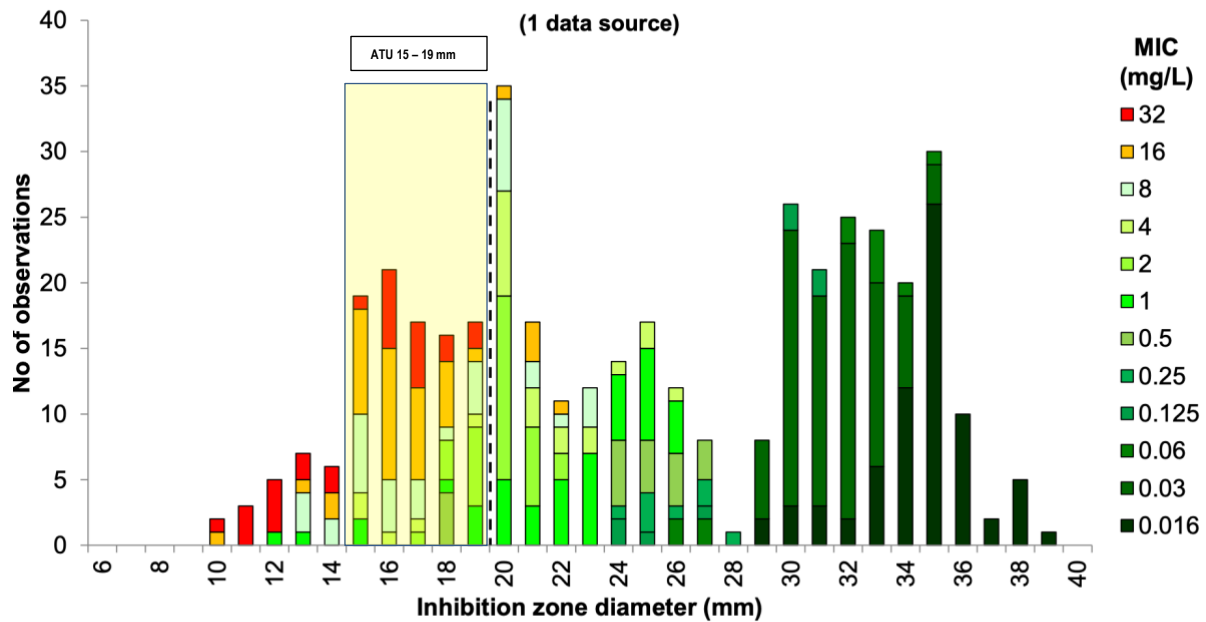
The ATU is a **warning** to laboratory staff set to avoid false susceptibility reports whereas some false resistance is accepted.

There is no EUCAST prescribed action. The warning may be ignored or acted upon. On occasion an alternative test may solve the issue but otherwise a decision must be taken to either report as tested with or without a warning or to refrain from reporting a formal S, I or R and instead add a comment to explain. On occasion, a wide ATU is needed but most ATUs are 1 – 2 mm (see figures below).

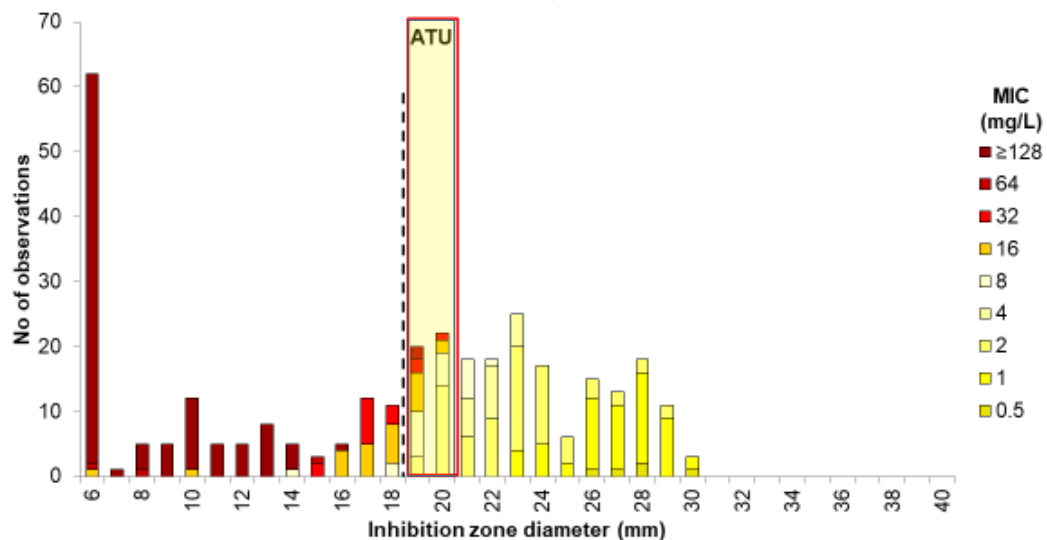
There are alternative actions for a result in the ATU:

1. Respect the warning and take measures to resolve the conundrum.
2. Ignore the warning and report as tested.

Meropenem-vaborbactam 20-10 µg vs. MIC
***Enterobacterales*, 104 isolates (412 correlates)**



Amoxicillin-clavulanic acid 20-10 µg vs MIC
***Enterobacterales*, 325 isolates**



Ad 1. Respect the warning - there are several alternatives (listed below) and the appropriateness of each will vary with the sample in which the organism was identified, the need for and importance of offering an interpretation (UTI vs. blood stream infection), the number of available therapeutic alternatives (as decided by the susceptibility pattern of the isolate) and whether the issue can be quickly resolved.

The individual laboratory may decide that results in the ATU should prompt laboratory technicians to consult with medical staff.

- **perform an alternative test** (perform an MIC, a PCR, a test to disclose the probable resistance mechanism or gene) – this is relevant when the alternative test is conclusive (PCR to detect a *mecA* or *mecC* gene in Staphylococci, a *vanA* or *vanB* gene in enterococci, a test to confirm the presence of an important enzyme).
- **report results in the ATU as “R” or “blank” with a comment.** If there are several good alternatives in the AST report this may be the easiest and safest option.
- **repeat the test** – this is only if there is reason to suspect a technical error.
- **discuss the results with the clinical colleague.**

Ad 2. Ignore the warning and interpret according to the result obtained (MIC or zone). Consider including a warning about the uncertainty of the result and/or interpretation. The ATU never interferes with the interpretation.

ATU warnings are listed in EUCAST Breakpoint Tables. They may change over time as breakpoints are updated.

A link which takes the reader to a document with graphs illustrating the use of ATUs is available next to the link to this document.